

Probability:-

□ Random variable:- is a mathematical variable whose value depend on the outcome of some uncertain Process.

→ Discrete Random variable Σ

→ Continuous Random variable $\int$

$$\bullet \Sigma P(X=x) = 1$$

$$\bullet 0 \leq P(X=x) \leq 1$$

↓
impossible

↓
certain

* Expected value / Mean

$$M = E(x) = \sum_x x \cdot P(X=x)$$

* Variance

$$\sigma^2 = \text{Var}(x) = E[(x-M)^2] = \sum_x (x-M)^2 \cdot P(X=x)$$

* standard deviation

$$\sigma = \sqrt{\text{Var}(x)}$$

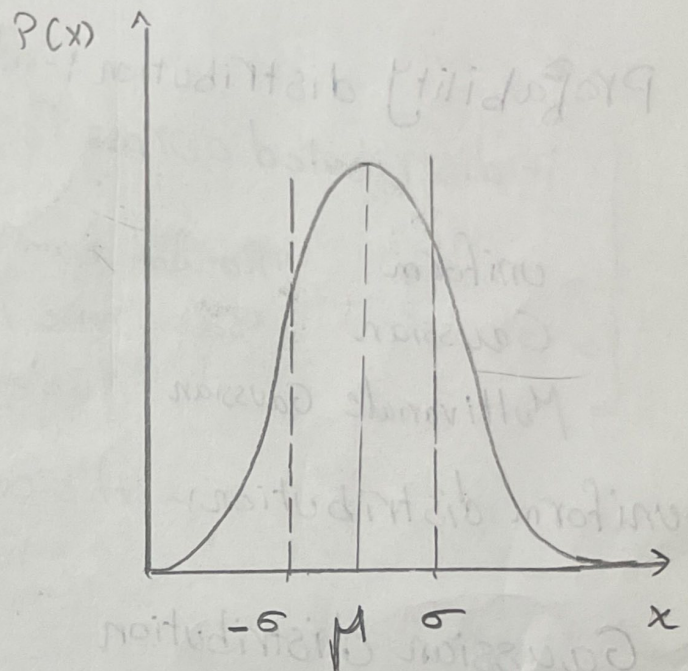
* Covariance

$$\text{Cov}(X, Y) = E[(X-M_x)(Y-M_y)]$$

⊕ increase x, y

⊖ $x \uparrow, y \downarrow$ or $x \downarrow, y \uparrow$

⊙ linear



low variance $\Rightarrow$ more consistent measurements

high variance $\Rightarrow$ more uncertainty

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* Conditional Probability,

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

Given

* Independence $\rightarrow P(A|B) = P(A)$
 $\rightarrow P(A \cap B) = P(A)P(B)$

* Total Probability

$$P(A) = \sum_i P(A|B_i)P(B_i)$$

Probability distribution:- describes how Probability is distributed across possible values

- Uniform "Random Sampling"
- Gaussian "Sensor noise / Kalman Filter"
- Multivariate Gaussian "Robot State Estimation"

→ uniform distribution:- The every value within an interval has the same Probability density

→ Gaussian distribution

$$f(x) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

$$f(\mu-a) = f(\mu+a)$$

$$\int f(x) dx = 1$$

$$\Rightarrow \mu_C = \frac{\sigma_B^2 \mu_A + \sigma_A^2 \mu_B}{\sigma_B^2 + \sigma_A^2}$$

$$\mu_C = \mu_A + \mu_B$$

$$\sigma_C^2 = \frac{1}{\frac{1}{\sigma_B^2} + \frac{1}{\sigma_A^2}}$$

$$\sigma_C^2 = \sigma_A^2 + \sigma_B^2$$

